

Solutions

Profit Max, Supply, Market Eq
Problem Set

Problem 1

$$(a) \quad f(m, c) = \min \left\{ \frac{m}{1/2}, \frac{c}{1} \right\} = \min \{ 2m, c \}$$

(b) $m_0 = 8 \Rightarrow$ The max amount of cups that can be produced is $q_{\max} = 16$

$$\Rightarrow C_{\text{sr}}(q, p_m, p_c) = \begin{cases} p_c q + 8p_m & 0 \leq q < 16 \\ [16p_c + 8p_m, \infty) & q = 16 \end{cases}$$

$$MC_{\text{sr}}(q, p_m, p_c) = \begin{cases} p_c & 0 \leq q < 16 \\ [p_c, \infty) & q = 16 \end{cases}$$

$$AC_{\text{sr}}(q, p_m, p_c) = \begin{cases} p_c + \frac{8p_m}{q} & 0 \leq q < 16 \\ [p_c + \frac{p_m}{2}, \infty) & q = 16 \end{cases}$$

$$AVC_{\text{sr}}(q, p_m, p_c) = \begin{cases} p_c & 0 \leq q < 16 \\ [p_c, \infty) & q = 16 \end{cases}$$

$$AFC_{\text{sr}}(q, p_m, p_c) = \begin{cases} \frac{8p_m}{q} & 0 \leq q < 16 \\ [p_m/2, \infty) & q = 16 \end{cases}$$

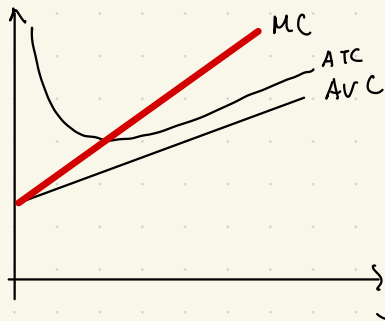
(c) Shutdown price: $\min AVC_{\text{sr}}$
 $\Rightarrow p^{\text{shutdown}} = p_c$

(d) This is the Long-Run Problem, with a fixed cost of $p_m \cdot 8$ (Remember, in the Long-Run there can be fixed costs but not sunk costs; Medina can choose to continue with the contract or not)

In the Long-Run, the shutdown decision is about profits $\Rightarrow$
Shutdown price: $\min AC$
 $\Rightarrow p_c + \frac{p_m}{2}$

Problem 2

$$\begin{aligned}
 (a) \quad MC(y) &= 2 + 2y \\
 ATC(y) &= \frac{k^2}{y} + 2 + y \\
 AVC(y) &= 2 + y \\
 AFC(y) &= \frac{k^2}{y}
 \end{aligned}$$



(b)

— Supply

$$p = 2 + 2y$$

$$\frac{p-2}{2} = y(p)$$

if $p > 2$,
 $\emptyset$ otherwise

$$(c) \quad Y^S(p) = J \left(\frac{p-2}{2} \right)$$

since they are identical J firms

$$\begin{aligned}
 (d) \quad Y^* &= Y^S(p^*) = Y^D(p^*) \\
 \Rightarrow J \left(\frac{p^*-2}{2} \right) &= A - Bp^* \\
 \Rightarrow p^* &= \frac{2(A+J)}{2B+J} \\
 Y^* &= \frac{(A-2B)J}{2B+J}
 \end{aligned}$$

(e) For $Y^* \geq 0$ we need $A > 2B$

$$\Leftrightarrow \frac{A}{2} > B$$

(f) Nothing because fixed costs are sunk in the short run, so they play no role in the firm's decision

(g) The zero-profit condition is found when $\min ATC$: this point is found when $MC(y^*) = ATC(y^*)$

$$\Rightarrow 2 + 2y^* = \frac{k^2}{y^*} + 2 + y^* \Rightarrow y^{*2} = k^2 \Rightarrow y^* = k$$

Thus individual production is $y^* = k$ at a price of $p^* = 2 + 2k \Rightarrow$ The number J^* of firms in the LR satisfies:

$$2 + 2k = \frac{2(A + J^*)}{2B + J^*}$$

$$\therefore J^* = \frac{A - 2B(1+k)}{k}$$

(h) A : higher willingness to pay $\Rightarrow \uparrow J^*$
 B is inversely proportional to the absolute value of the price elasticity of demand.
 $B \uparrow \Rightarrow J^* \downarrow$, i.e. less elastic, less firms.
 k : higher fixed costs, less firms.

(i) Assuming free entry, since the firms are identical, then the long run supply curve is horizontal at a price equal to $2 + 2k$

Problem 3

Firm

$$(a) \max_L 4\sqrt{L} - 1 - wL$$

$$(b) \text{Optimality } 4 \cdot MP_L = w \Rightarrow \frac{2}{\sqrt{L}} = w$$

$$MP_L = \frac{1}{2\sqrt{L}}$$

$$\therefore L^4 = \frac{4}{w^2}$$

$$(c) \text{Profits} = 4\sqrt{4/w^2} - 1 - w \frac{4}{w^2} = \frac{8}{w} - \frac{4}{w} - 1$$

$$= \frac{4-w}{w}$$

$$(d) L_D(w) = \frac{400}{w^2} \text{ since firms are identical}$$

Market

$$(i) L_D(w^*) = L_S(w^*) \Rightarrow \frac{400}{w^2} = \frac{200,000 w^*}{1+w^*}$$

$$w^* = 0.13128$$

$$L^* = 23209.2$$

Worker

$$(e) x = w(200-h) \Rightarrow x + wh = 200w$$

$$(f) \max_{x,h} x^{1/2} + h^{1/2}$$

$$\text{s.t. } x + wh = 200w$$

$$MRS = \frac{\frac{1}{2\sqrt{x}}}{\frac{1}{2\sqrt{h}}} = \frac{\sqrt{h}}{\sqrt{x}}$$

$$\text{Optimality cond. } \frac{\sqrt{h}}{\sqrt{x}} = \frac{1}{w} \Rightarrow \sqrt{h} w = \sqrt{x}$$

$$(g) \text{ If } w=1 \Rightarrow h^* = x^* \Rightarrow x^* + h^* = 200$$

$$\Rightarrow x^* = h^* = 100$$

$$(h) \frac{h^*}{x^*} = \frac{1}{w^2} \Rightarrow h^* w^2 = x^*$$

$$\Rightarrow h^* (w^2 + w) = 200w$$

$$\Rightarrow h^* = \frac{200}{1+w}$$

$$l^* = 200 - h^* = 200 - \frac{200}{1+w} = \frac{200 + 200w - 200}{1+w}$$

$$l^*(w) = \frac{200w}{1+w}$$

$$(i) = L_S(w) = \frac{200,000 w}{1+w}$$